

Introduction to Radiobiological Models

A short story



Souvik Das

Resident Medical Physicist
Tata Memorial Hospital, Mumbai

Table of Contents

- 1 Linear Quadratic (LQ) Model
- 2 Nominal Standard Dose (NSD)
- 3 Cumulative Radiation Effect (CRE)
- 4 Time-Dose-Fractionation (TDF)
- 5 Practical Applications

Table of Contents

- 1 Linear Quadratic (LQ) Model
- 2 Nominal Standard Dose (NSD)
- 3 Cumulative Radiation Effect (CRE)
- 4 Time-Dose-Fractionation (TDF)
- 5 Practical Applications

- The target of radiation for the purpose of therapy is the **DNA**.

- The target of radiation for the purpose of therapy is the **DNA**.
- Post-irradiation **cell survival** as a function of radiation dose can be **biophysically modeled** in several different ways.

- The target of radiation for the purpose of therapy is the **DNA**.
- Post-irradiation **cell survival** as a function of radiation dose can be **biophysically modeled** in several different ways.
- **Poisson statistics** are used for the basis for survival equations.

Mathematical Modeling

Poisson Statistics

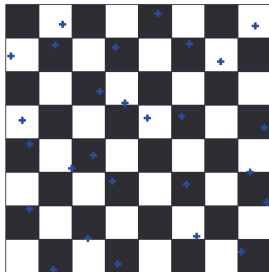


Figure: If it rains on a chessboard, and an average of x drops hits each square, How many squares are dry?

Mathematical Modeling

Poisson Statistics

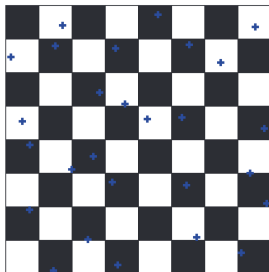


Figure: If it rains on a chessboard, and an average of x drops hits each square, How many squares are dry?

- Poisson statistics describe a large number of random events happening to a large number of subjects, averaging out to a small number of events per subject.

Mathematical Modeling

Poisson Statistics

- If x is the average number of drops per square, then the probability of a square being hit by k drops is given by:

$$P(k) = \frac{x^k e^{-x}}{k!}$$

Mathematical Modeling

Poisson Statistics

- If x is the average number of drops per square, then the probability of a square being hit by k drops is given by:

$$P(k) = \frac{x^k e^{-x}}{k!}$$

- The probability of a square being dry is given by:

$$P(0) = e^{-x}$$

Mathematical Modeling

Poisson Statistics

- If x is the average number of drops per square, then the probability of a square being hit by k drops is given by:

$$P(k) = \frac{x^k e^{-x}}{k!}$$

- The probability of a square being dry is given by:

$$P(0) = e^{-x}$$

- The probability of a square being wet is given by:

$$P(\text{wet}) = 1 - e^{-x}$$

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
 - At an average of x lethal hits per cell:

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
 - At an average of x lethal hits per cell:
 - e^{-x} cells survive (no hits).

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
 - At an average of x lethal hits per cell:
 - e^{-x} cells survive (no hits).
 - $1 - e^{-x}$ cells die (at least 1 hit).

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
 - At an average of x lethal hits per cell:
 - e^{-x} cells survive (no hits).
 - $1 - e^{-x}$ cells die (at least 1 hit).
 - Based on this equation,

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
 - At an average of x lethal hits per cell:
 - e^{-x} cells survive (no hits).
 - $1 - e^{-x}$ cells die (at least 1 hit).
 - Based on this equation,
 - 1 hit kills 63% of cells. $SF = 0.37$

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
 - At an average of x lethal hits per cell:
 - e^{-x} cells survive (no hits).
 - $1 - e^{-x}$ cells die (at least 1 hit).
 - Based on this equation,
 - 1 hit kills 63% of cells. $SF = 0.37$
 - 2 hits kills 86% of cells. $SF = 0.14$

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
 - At an average of x lethal hits per cell:
 - e^{-x} cells survive (no hits).
 - $1 - e^{-x}$ cells die (at least 1 hit).
 - Based on this equation,
 - 1 hit kills 63% of cells. $SF = 0.37$
 - 2 hits kills 86% of cells. $SF = 0.14$
 - 3 hits kills 95% of cells. $SF = 0.05$

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
 - At an average of x lethal hits per cell:
 - e^{-x} cells survive (no hits).
 - $1 - e^{-x}$ cells die (at least 1 hit).
 - Based on this equation,
 - 1 hit kills 63% of cells. $SF = 0.37$
 - 2 hits kills 86% of cells. $SF = 0.14$
 - 3 hits kills 95% of cells. $SF = 0.05$
 - 4 hits kills 98% of cells. $SF = 0.02$

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
 - At an average of x lethal hits per cell:
 - e^{-x} cells survive (no hits).
 - $1 - e^{-x}$ cells die (at least 1 hit).
 - Based on this equation,
 - 1 hit kills 63% of cells. $SF = 0.37$
 - 2 hits kills 86% of cells. $SF = 0.14$
 - 3 hits kills 95% of cells. $SF = 0.05$
 - 4 hits kills 98% of cells. $SF = 0.02$
 - 5 hits kills 99% of cells. $SF = 0.01$

Poisson Distribution

Cell Survival

- The Poisson model is used for surviving fraction (**SF**) of cells:
 - At an average of x lethal hits per cell:
 - e^{-x} cells survive (no hits).
 - $1 - e^{-x}$ cells die (at least 1 hit).
 - Based on this equation,
 - 1 hit kills 63% of cells. $SF = 0.37$
 - 2 hits kills 86% of cells. $SF = 0.14$
 - 3 hits kills 95% of cells. $SF = 0.05$
 - 4 hits kills 98% of cells. $SF = 0.02$
 - 5 hits kills 99% of cells. $SF = 0.01$

Note

D_0 is defined as the radiation dose resulting in 37% survival. This is assumed to be equal to the radiation dose necessary to cause one **lethal** hit per cell.

Poisson Distribution

Tumor Control Probability (TCP)

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:

Poisson Distribution

Tumor Control Probability (TCP)

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
 - At an average of x surviving tumor cells per patient:

Poisson Distribution

Tumor Control Probability (TCP)

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
 - At an average of x surviving tumor cells per patient:
 - e^{-x} cells survive (no hits).

Poisson Distribution

Tumor Control Probability (TCP)

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
 - At an average of x surviving tumor cells per patient:
 - e^{-x} cells survive (no hits).
 - $1 - e^{-x}$ cells die (at least 1 hit).

Poisson Distribution

Tumor Control Probability (TCP)

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
 - At an average of x surviving tumor cells per patient:
 - e^{-x} cells survive (no hits).
 - $1 - e^{-x}$ cells die (at least 1 hit).
 - Based on this equation,

Poisson Distribution

Tumor Control Probability (TCP)

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
 - At an average of x surviving tumor cells per patient:
 - e^{-x} cells survive (no hits).
 - $1 - e^{-x}$ cells die (at least 1 hit).
 - Based on this equation,
 - 1 tumor cell per pt.: $TCP = 0.37$

Poisson Distribution

Tumor Control Probability (TCP)

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
 - At an average of x surviving tumor cells per patient:
 - e^{-x} cells survive (no hits).
 - $1 - e^{-x}$ cells die (at least 1 hit).
 - Based on this equation,
 - 1 tumor cell per pt.: $TCP = 0.37$
 - 2 tumor cell per pt.: $TCP = 0.14$

Poisson Distribution

Tumor Control Probability (TCP)

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
 - At an average of x surviving tumor cells per patient:
 - e^{-x} cells survive (no hits).
 - $1 - e^{-x}$ cells die (at least 1 hit).
 - Based on this equation,
 - 1 tumor cell per pt.: $TCP = 0.37$
 - 2 tumor cell per pt.: $TCP = 0.14$
 - 3 tumor cells per pt.: $TCP = 0.05$

Poisson Distribution

Tumor Control Probability (TCP)

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
 - At an average of x surviving tumor cells per patient:
 - e^{-x} cells survive (no hits).
 - $1 - e^{-x}$ cells die (at least 1 hit).
 - Based on this equation,
 - 1 tumor cell per pt.: $TCP = 0.37$
 - 2 tumor cell per pt.: $TCP = 0.14$
 - 3 tumor cells per pt.: $TCP = 0.05$
 - 4 tumor cells per pt.: $TCP = 0.02$

Poisson Distribution

Tumor Control Probability (TCP)

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
 - At an average of x surviving tumor cells per patient:
 - e^{-x} cells survive (no hits).
 - $1 - e^{-x}$ cells die (at least 1 hit).
 - Based on this equation,
 - 1 tumor cell per pt.: $TCP = 0.37$
 - 2 tumor cell per pt.: $TCP = 0.14$
 - 3 tumor cells per pt.: $TCP = 0.05$
 - 4 tumor cells per pt.: $TCP = 0.02$
 - 5 tumor cells per pt.: $TCP = 0.01$

Poisson Distribution

Tumor Control Probability (TCP)

- The Poisson model is used for Tumor Control Probability (**TCP**) of cells:
 - At an average of x surviving tumor cells per patient:
 - e^{-x} cells survive (no hits).
 - $1 - e^{-x}$ cells die (at least 1 hit).
 - Based on this equation,
 - 1 tumor cell per pt.: $TCP = 0.37$
 - 2 tumor cell per pt.: $TCP = 0.14$
 - 3 tumor cells per pt.: $TCP = 0.05$
 - 4 tumor cells per pt.: $TCP = 0.02$
 - 5 tumor cells per pt.: $TCP = 0.01$

Note

To achieve a certain TCP, you should aim for a tumor cell survival of $(1 - TCP)$.

Target theory

Single-Target, Single-Hit Model

- It assumes that each cell has a **single critical target**, and that a single hit to this target is sufficient to cause cell death.

Target theory

Single-Target, Single-Hit Model

- It assumes that each cell has a **single critical target**, and that a single hit to this target is sufficient to cause cell death.
- The survival fraction (SF) is given by:

$$SF = e^{-D/D_0}$$

where D is the dose and D_0 is the dose required to reduce the surviving fraction to 37%.

Target theory

Single-Target, Single-Hit Model

- It assumes that each cell has a **single critical target**, and that a single hit to this target is sufficient to cause cell death.
- The survival fraction (SF) is given by:

$$SF = e^{-D/D_0}$$

where D is the dose and D_0 is the dose required to reduce the surviving fraction to 37%.

- This type of survival curve is seen for mammalian cells which are irradiated with **high LET** radiation such as alpha particles or carbon ions; very **radiation-sensitive** cells like lymphocytes and bone marrow cells; cells that have major defects in DNA doublestrand break repair; are synchronized in M phase; and for cells that have DNA double-strand break repair inhibited by chemical or gene knockdown.

Target theory

Multitarget, Single-Hit Model

- This model assumes that each cell has multiple independent targets, all of which must be hit to kill the cell.

Target theory

Multitarget, Single-Hit Model

- This model assumes that each cell has multiple independent targets, all of which must be hit to kill the cell.
- For a single dose D , the survival fraction (SF) is given by:

$$SF = 1 - (1 - e^{-D/D_0})^n$$

where D is the dose, D_0 is the dose required to reduce the surviving fraction to 37%, and n is the number of targets.

Target theory

Multitarget, Single-Hit Model

- This model assumes that each cell has multiple independent targets, all of which must be hit to kill the cell.
- For a single dose D , the survival fraction (SF) is given by:

$$SF = 1 - (1 - e^{-D/D_0})^n$$

where D is the dose, D_0 is the dose required to reduce the surviving fraction to 37%, and n is the number of targets.

- This type of survival curve is seen for mammalian cells which are irradiated with **low LET** radiation such as X-rays or gamma rays; cells that are **radiation-resistant** like fibroblasts and epithelial cells; cells that have a normal DNA double-strand break repair; and for cells that are synchronized in G1 or S phase.

Target theory

Multitarget, Single-Hit Model

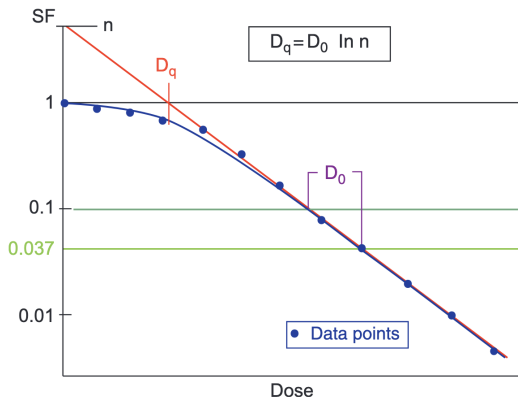


Figure: The single-hit, multitarget survival curve. Notice that it is curved at low doses and straight (e.g., survival is an exponential function of dose) at high doses. This allows you to calculate D_0 by taking measurements in the high-dose region

Target theory

Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:

Target theory

Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:
 - D_0 is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

Target theory

Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:
 - D_0 is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

Important to note

Do not measure D_0 at $SF = 0.37$ if the survival curve has a shoulder. Remember that D_0 applies to the high-dose portion of the curve in that case! Instead, measure the difference in dose between $SF = 0.1$ and $SF = 0.037$.

Target theory

Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:
 - D_0 is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

Important to note

Do not measure D_0 at $SF = 0.37$ if the survival curve has a shoulder. Remember that D_0 applies to the high-dose portion of the curve in that case! Instead, measure the difference in dose between $SF = 0.1$ and $SF = 0.037$.

- D_0 is a measure of the cell's inherent radiation sensitivity. Most D_0 values are somewhere around 1 Gy.

Target theory

Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:
 - D_0 is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

Important to note

Do not measure D_0 at $SF = 0.37$ if the survival curve has a shoulder. Remember that D_0 applies to the high-dose portion of the curve in that case! Instead, measure the difference in dose between $SF = 0.1$ and $SF = 0.037$.

- D_0 is a measure of the cell's inherent radiation sensitivity. Most D_0 values are somewhere around 1 Gy.
- $D_{10} = 2.3 \times D_0 =$ a dose that will reduce surviving fraction by tenfold (i.e., from $SF = 0.1$ to $SF = 0.01$).

Target theory

Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:
 - D_0 is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

Important to note

Do not measure D_0 at $SF = 0.37$ if the survival curve has a shoulder. Remember that D_0 applies to the high-dose portion of the curve in that case! Instead, measure the difference in dose between $SF = 0.1$ and $SF = 0.037$.

- D_0 is a measure of the cell's inherent radiation sensitivity. Most D_0 values are somewhere around 1 Gy.
- $D_{10} = 2.3 \times D_0 =$ a dose that will reduce surviving fraction by tenfold (i.e., from $SF = 0.1$ to $SF = 0.01$).
- The low-dose portion of the curve is known as the “shoulder”:

Target theory

Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:
 - D_0 is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

Important to note

Do not measure D_0 at $SF = 0.37$ if the survival curve has a shoulder. Remember that D_0 applies to the high-dose portion of the curve in that case! Instead, measure the difference in dose between $SF = 0.1$ and $SF = 0.037$.

- D_0 is a measure of the cell's inherent radiation sensitivity. Most D_0 values are somewhere around 1 Gy.
- $D_{10} = 2.3 \times D_0 =$ a dose that will reduce surviving fraction by tenfold (i.e., from $SF = 0.1$ to $SF = 0.01$).
- The low-dose portion of the curve is known as the “shoulder”:
 - D_q tells you how wide the shoulder is.

Target theory

Multitarget, Single-Hit Model

- The high-dose portion of the curve is actually a straight line:
 - D_0 is defined as the additional dose required to reduce surviving fraction to 37% of what it was before.

Important to note

Do not measure D_0 at $SF = 0.37$ if the survival curve has a shoulder. Remember that D_0 applies to the high-dose portion of the curve in that case! Instead, measure the difference in dose between $SF = 0.1$ and $SF = 0.037$.

- D_0 is a measure of the cell's inherent radiation sensitivity. Most D_0 values are somewhere around 1 Gy.
- $D_{10} = 2.3 \times D_0$ = a dose that will reduce surviving fraction by tenfold (i.e., from $SF = 0.1$ to $SF = 0.01$).
- The low-dose portion of the curve is known as the “shoulder”:
 - D_q tells you how wide the shoulder is.
 - D_q is a measure of the cell's repair capacity. More repair means a larger D_q and a larger shoulder

Single-Hit, Multitarget Model

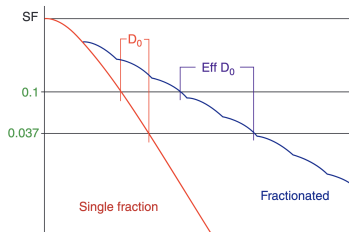
Advantages and Disadvantages

Fractionated Radiation and Effective D_0

- A fractionated radiation survival curve has a shallower slope compared to a single-fraction survival curve.

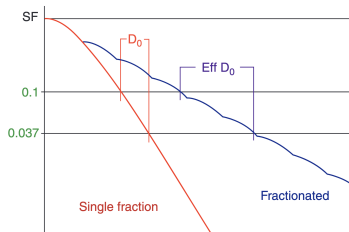
Fractionated Radiation and Effective D_0

- A fractionated radiation survival curve has a shallower slope compared to a single-fraction survival curve.
- Assuming complete repair in between fractions, the shoulder is repeated with each fraction.



Fractionated Radiation and Effective D_0

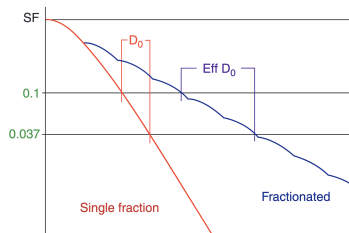
- A fractionated radiation survival curve has a shallower slope compared to a single-fraction survival curve.
- Assuming complete repair in between fractions, the shoulder is repeated with each fraction.
- D_0 versus effective D_0 . When radiation is fractionated, you need much more dose to achieve the same amount of cell kill. Therefore, effective D_0 is always larger than D_0



Fractionated Radiation and Effective D_0

Continue

- The slope of a fractionated survival curve is called effective D_0 .



Fractionated Radiation and Effective D_0

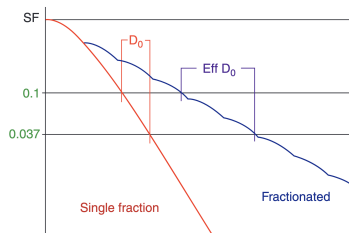
Continue

- The slope of a fractionated survival curve is called effective D_0 .
- For a single fraction of D Gy with a surviving fraction of SF_D :

$$SF_D = e^{-D/D_0}$$

$$SF_D = e^{-D/\text{effective } D_0}$$

$$\text{effective } D_0 = -\frac{D}{\ln(SF_D)}$$



Fractionated Radiation and Effective D_0

Continue

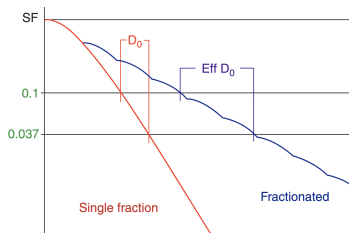
- The slope of a fractionated survival curve is called effective D_0 .
- For a single fraction of D Gy with a surviving fraction of SF_D :

$$SF_D = e^{-D/D_0}$$

$$SF_D = e^{-D/\text{effective } D_0}$$

$$\text{effective } D_0 = -\frac{D}{\ln(SF_D)}$$

- After x fractions to total dose xD Gy:
 $SF_x^D = e^{-xD/\text{effective } D_0}$



Linear-Quadratic (LQ, Alpha-Beta) Model

- The linear-quadratic model was developed after the in vitro observation that DNA damage follows a linear-quadratic relationship with dose D .

Linear-Quadratic (LQ, Alpha-Beta) Model

- The linear-quadratic model was developed after the in vitro observation that DNA damage follows a linear-quadratic relationship with dose D .
- Lethal DNA aberrations $= \alpha D + \beta D^2 = \text{Cell Kill}$

$$SF_D = e^{-(\alpha D + \beta D^2)}$$

Linear-Quadratic (LQ, Alpha-Beta) Model

- The linear-quadratic model was developed after the in vitro observation that DNA damage follows a linear-quadratic relationship with dose D .
- Lethal DNA aberrations = $\alpha D + \beta D^2$ = Cell Kill

$$SF_D = e^{-(\alpha D + \beta D^2)}$$

- Unlike the single-hit model, the LQ model accounts for two different types of lethal hits which are based on known molecular mechanisms of DNA damage:

Linear-Quadratic (LQ, Alpha-Beta) Model

- The linear-quadratic model was developed after the in vitro observation that DNA damage follows a linear-quadratic relationship with dose D .
- Lethal DNA aberrations $= \alpha D + \beta D^2 = \text{Cell Kill}$

$$SF_D = e^{-(\alpha D + \beta D^2)}$$

- Unlike the single-hit model, the LQ model accounts for two different types of lethal hits which are based on known molecular mechanisms of DNA damage:
 - Single-hit kill (α) is unrepairable damage and is **independent** of fractionation or dose rate. This corresponds to single-hit and **intra**-track accumulated damage.

Linear-Quadratic (LQ, Alpha-Beta) Model

- The linear-quadratic model was developed after the in vitro observation that DNA damage follows a linear-quadratic relationship with dose D .
- Lethal DNA aberrations $= \alpha D + \beta D^2 = \text{Cell Kill}$

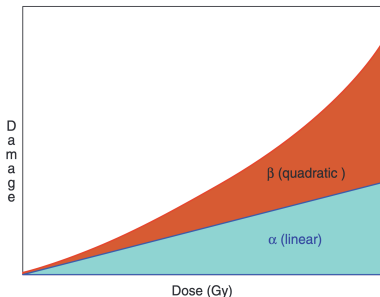
$$SF_D = e^{-(\alpha D + \beta D^2)}$$

- Unlike the single-hit model, the LQ model accounts for two different types of lethal hits which are based on known molecular mechanisms of DNA damage:
 - Single-hit kill (α) is unrepairable damage and is **independent** of fractionation or dose rate. This corresponds to single-hit and **intra**-track accumulated damage.
 - Multi-hit kill (β) is repairable damage and is **dependent** on fractionation or dose rate. This corresponds to double-hit and **inter**-track accumulated damage.

Linear-Quadratic (LQ, Alpha-Beta) Model

Continue:

- The α/β ratio is the **dose** at which α kill and β kill are equal.
 - Low α/β ratio (“high repair”) tissues are relatively resistant at small fraction size and relatively sensitive at large fraction size.
 - High α/β ratio (“low repair”) tissues are relatively sensitive at small fraction size and relatively resistant at large fraction size.



Linear-Quadratic (LQ, Alpha-Beta) Model

Alpha-Beta Model and Dose Fractionation

- LQ model is widely used in radiotherapy to predict the effects of different fractionation schemes on both tumor and normal tissues.
- Suppose we have n no of fractions of dose d each, then the total dose $D = nd$. The surviving fraction after n fractions is given by:

$$\begin{aligned} SF_n &= e^{-n(\alpha d + \beta d^2)} \\ \Rightarrow -\frac{1}{nd} &= \frac{\alpha}{\ln(SF_n)} + \frac{\beta d}{\ln(SF_n)} \end{aligned}$$

- Plot of dose per fraction d versus $-\frac{1}{nd}$ gives a straight line with slope $\beta/\ln(SF_n)$ and y-intercept $\ln(SF_n)$. The ratio of intercept to slope gives α/β .
- Such values are obtained for a variety of tissues and tumors, and are used to guide clinical decision-making regarding fractionation schemes.

Linear-Quadratic (LQ, Alpha-Beta) Model

α/β Ratios of Tissues and Tumor

- Most acute-reacting tissues are believed to have an α/β ratio ≈ 10 .
- Most late-reacting tissues are believed to have an α/β ratio ≈ 3 or less.
- CNS tissue (brain, cord) is believed to have an α/β ratio $\approx 1-2$.
- Most tumors are believed to have an α/β ratio ≥ 10 .
- Some low α/β tumors may have an α/β ratio $\approx 1.5-4$. Breast and Prostate are the “classic” low α/β tumors.

Linear-Quadratic (LQ, Alpha-Beta) Model

Biologically Effective Dose

- One advantage of the linear-quadratic model is that it is easy to calculate effective doses for different fraction sizes.

Definition

Biologically effective dose (BED) is an extrapolated dose given over infinitely many fractions.

- For n fractions of d dose per fraction:

$$\begin{aligned}\text{BED}_{\alpha/\beta} &= nd \left(1 + \frac{d}{\alpha/\beta} \right) \\ &= D \left(1 + \frac{d}{\alpha/\beta} \right)\end{aligned}$$

- $\text{BED} = \text{total physical dose} \times \text{relative effectiveness of that dose.}$

Linear-Quadratic (LQ, Alpha-Beta) Model

Practical Applications

Problem 1

If a treatment is planned for 70 Gy/ 35 fractions, but due to departmental constraints, we need to complete it in 25 fractions. What is the new dose per fraction for same tumor control with $\alpha/\beta = 10$?

Linear-Quadratic (LQ, Alpha-Beta) Model

Equivalent Dose in 2 Gy fractions (EQD₂)

- EQD₂ is the dose in 2 Gy fractions that would produce the same biological effect as the actual treatment.
- For n fractions of d dose per fraction:

$$\text{EQD}_2 = \frac{BED}{1 + \frac{2}{\alpha/\beta}}$$
$$\Rightarrow \text{EQD}_2 = D \left(\frac{d + \alpha/\beta}{2 + \alpha/\beta} \right)$$

Linear-Quadratic (LQ, Alpha-Beta) Model

Practical Applications

Problem 2

You want to treat a lung cancer to 60 Gy in 2 Gy fractions, but the first five fractions are given at 3 Gy/fx due to SVC syndrome. Assuming $\alpha/\beta = 3$, how much additional dose should you deliver at 2 Gy per fraction?

Table of Contents

- 1 Linear Quadratic (LQ) Model
- 2 Nominal Standard Dose (NSD)
- 3 Cumulative Radiation Effect (CRE)
- 4 Time-Dose-Fractionation (TDF)
- 5 Practical Applications

Table of Contents

- 1 Linear Quadratic (LQ) Model
- 2 Nominal Standard Dose (NSD)
- 3 Cumulative Radiation Effect (CRE)
- 4 Time-Dose-Fractionation (TDF)
- 5 Practical Applications

Table of Contents

- 1 Linear Quadratic (LQ) Model
- 2 Nominal Standard Dose (NSD)
- 3 Cumulative Radiation Effect (CRE)
- 4 Time-Dose-Fractionation (TDF)**
- 5 Practical Applications

Table of Contents

- 1 Linear Quadratic (LQ) Model
- 2 Nominal Standard Dose (NSD)
- 3 Cumulative Radiation Effect (CRE)
- 4 Time-Dose-Fractionation (TDF)
- 5 Practical Applications